

Notes on lectures 0, 1, and 2

LECTURE 0

In this course we will cover a number of different types of optimization. The first of these is *unconstrained optimization*, where we place no constraints on our *decision variable* $x \in \mathbb{R}^n$. The counterpart to this is *constrained optimization*, in which we restrict x to some *feasible region* $X \in \mathbb{R}^n$.

Another type of optimization is *minimax optimization* in which theorems take the form

$$\max_{x \in X} \min_{y \in Y} f(x, y) = \min_{y \in Y} \max_{x \in X} f(x, y).$$

Wikipedia tells me that John von Neumann was the first to produce a theorem of this type (back in 1928), and that his theorem marked the start of game theory.

Stochastic optimization concerns methods that involve random variables. *Nonsmooth optimization* concerns nonsmooth functions. This area seems to stand apart from traditional optimization and its techniques, which seem to deal with smooth functions. *Nonconvex optimization* deals with nonconvex functions, the optimization of which cannot make use of the results relied upon in convex optimization. Standard techniques for convex optimization, such as gradient descent, may fail when applied to nonconvex optimization problems.

The notes seem to make clear that many concepts and results from linear algebra may be helpful to us here. One concept from linear algebra is the *rank* of a matrix A , which gives the number of linearly independent rows (or, equivalently, columns) of A . An *eigenvalue* of a square matrix A is some scalar λ for which $Av = \lambda v$. The notion of *singular value* of a matrix is closely tied to that of an eigenvalue. If a square matrix A is diagonalizable, then it has an *eigendecomposition*; in other words, it can be written as $A = V\Lambda V^\top$, where Λ is a diagonal matrix that contains the eigenvalues of A along its diagonal. Similarly, *every* matrix A has a *singular value decomposition*, and hence may be written as $A = U\Sigma V^\top$, where Σ is a diagonal matrix that contains the singular values of A along its diagonal.

Another notion is that of the *inner product* $x^\top y$ of two vectors x and y (having equal dimension). When $x^\top y = 0$, we say that x and y are *orthogonal*, and can denote this by $x \perp y$. When we take the inner product of a vector x with itself, we get its *Euclidean* or ℓ_2 norm, which we denote by $\|x\|_2$, and which represents the *magnitude* or *length* of x .

From probability, *random variables* are used to represent outcomes of random events, such as the result of a coin toss. *Expectation* is another concept from probability, and can be seen as giving the average result from a large number of repetitions of the same random experiment, such as the rolling of a die. The *law of large numbers* says, as our number of repetitions of a random experiment increases towards infinity, our average result approaches the expectation, also known as the *expected value*.

Our task with optimization is to find a *best* decision; this implies that there may be more than one. It also implies that there may be more than one meaning of *best*. However, for us, *best* will typically mean global minimum or maximum.

The examples in the second half of the slide deck are meant to convey the idea that, when it comes to formulating an optimization problem, there is no “one-size-fits-all” solution; it seems rather to be the case

that the formulation should be tailored to the problem. Another observation is that simple reframing of a problem can necessitate a change in the approach to solve it. A linear model $y_i = \theta^\top x_i$ was chosen for the task of predicting Chloe’s rating, while changing the task to a binary classification problem led to the use of a logistic model instead—all the while the input data remained the same.

LECTURE 1 MATH REVIEW

There are a few notions from set theory which may be helpful to us here. One of these is the *Cartesian product* $X_1 \times X_2$ of two sets X_1 and X_2 . The elements of $X_1 \times X_2$ take the form (x_1, x_2) , where $x_1 \in X_1$ and $x_2 \in X_2$. We can also extend this to an arbitrary number of sets, in which case we will have something like

$$X_1 \times X_2 \times \dots \times X_n = \{(x_1, x_2, \dots, x_n) \mid x_1 \in X_1, x_2 \in X_2, \dots, x_n \in X_n\}.$$

The *infimum* of a set S represents the “greatest lower bound” of S . To make this more concrete, let’s say that $S = \{2, 3, 4\}$ is a subset of a larger set $P = \{1, 2, 3, 4, 5\}$. In this case, we can say that any element of P that is less than or equal to every element in S is a *lower bound* of S ; the infimum of S (denoted $\inf S$) will be the greatest of all these lower bounds. In our example, the lower bounds of S are $1, 2 \in P$, and the greater of these, 2, is the infimum of S (i.e., $\inf S = 2$). The *supremum*, on the other hand, represents the “least upper bound” of a set. In our example, we have the two upper bounds $4, 5 \in P$; the lesser of these, 4, is the supremum of S , and we can write $\sup S = 4$.

It is worth noting that the infimum or supremum of a set is not always an element of that set. In our example, both the infimum and supremum of S happened to be elements of S ; but this is not always the case. For example, in the case of the interval $[0, 1)$, we have $\inf[0, 1) = 0 \in [0, 1)$ while $\sup[0, 1) = 1 \notin [0, 1)$.

[Kun86] is a book on set theory I found at the bottom of the Wikipedia page for set theory.

In linear algebra we talk about vectors (for one thing) and operations on them. One thing we can do with two vectors x and y (having equal dimension) is “combine” them as $z = ax + by$ (where a and b are scalars) to obtain a new vector z which is a *linear combination* of x and y . Along with this concept of linear combination comes the notion of a *subspace*. A subspace is a set containing vectors that, when added together or multiplied by a scalar, produce new vectors that likewise belong to the set; for this reason a subspace is understood to be “closed” under addition and scalar multiplication. Another concept related to the idea of linear combination is *linear independence*. A set of vectors are *linearly independent* if the only way to make them sum to 0 is by having each scalar (that is part of their linear combination) be 0. This concept is closely tied to the notion of *dimension* of a subspace: the size of the largest linearly independent set that can be constructed from the vectors of a subspace gives the dimension of that subspace.

Putting it into my own words as best I can, norms provide us with notions of “size” for vectors and matrices. A norm is a function which must satisfy certain properties; one of the consequences of these properties is that the value of a norm is always nonnegative. The *Cauchy-Schwarz inequality* relates the inner product of two vectors x and y to the product of their norms in the following way:

$$x^\top y = \|x\| \|y\| \cos \theta.$$

Similarly, we have

$$\cos \theta = \frac{x^\top y}{\|x\| \|y\|},$$

where θ is the angle between x and y . We see that the equality $x^\top y = \|x\| \|y\|$ occurs whenever $\cos \theta = 1$; or, in other words, whenever the angle between the vectors $\theta = 0$ (the vectors are parallel).

The concept of an eigenvalue was discussed above. The largest eigenvalue of a matrix A is sometimes denoted $\lambda_{\max}(A)$. The magnitude of $\lambda_{\max}(A)$ is denoted $\rho(A)$ and is called the *spectral radius* of A .

The ℓ_2 norm of a matrix A is denoted $\|A\|_2$, and its definition

$$\|A\|_2 = \sup_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2}$$

shows in a way that it represents the greatest amount by which it changes the size of some vector x . Now, if $A \in \mathbb{R}^{m \times n}$, then this would seem to imply that we would need to consider every $x \in \mathbb{R}^n$ in order to find $\|A\|_2$. However, it turns out that $\|A\|_2 = \sigma_1$, the first singular value of A . Thus, one way to compute $\|A\|_2$ is to compute the singular value decomposition $A = U\Sigma V^\top$; then, $\|A\|_2 = \sigma_1 = \Sigma_{11}$.

Like the ℓ_2 norm, the *Frobenius norm* $\|A\|_F$ can also be written in terms of the singular values of A as $\|A\|_F = \sqrt{\sum_{i=1}^p \sigma_i^2}$; and likewise for the *nuclear norm* $\|A\|_* = \sum_{i=1}^p \sigma_i$.

Recall that the ℓ_2 norm $\|x\|_2$ of a vector x is equal to $\sqrt{x^\top x}$, the square root of its inner product with itself. Thus, $\|x\|_2^2 = x^\top x$, and we can almost see $x^\top x$ as a “square” of x . Similarly, $x^\top x A x$ is like the linear algebra analogue of ax^2 for scalars a and x . Thus in linear algebra too we have a notion of a “quadratic,” for a symmetric matrix $A = A^\top$ we have the following “quadratic-form bounds”:

$$\lambda_{\min}(A)\|x\|^2 \leq x^\top A x \leq \lambda_{\max}(A)\|x\|^2.$$

When $\lambda_{\min} \geq 0$, we say that A is *positive semi-definite*.

The *condition number* $\kappa(A)$ of a matrix A provides an idea of how much a slight change in a vector x may affect the vector Ax . The results of computations involving matrices with large condition numbers can be unreliable, and the condition number of a matrix is therefore an important measure in the area of numerical methods.

[Bea73] is an old textbook—also grabbed off Wikipedia—on linear algebra.

A function $f(x)$ is said to be $\mathcal{O}(g(x))$ (read “big Oh of $g(x)$ ”) if there is some point n_0 after which $f(n)$ is bounded above by $\alpha g(x)$ for some constant α . This means that we can find some value α such that $\alpha g(x)$ grows asymptotically faster than $f(x)$. If, on the other hand, we can find some value α that makes $\alpha g(x)$ grow asymptotically slower than $f(x)$, then in this case $f(x)$ is said to be $\Omega(g(x))$ (read “big Omega of $g(x)$ ”), and $g(x)$ serves as a lower bound on $f(x)$.

The definitions for o (“small Oh”) and ω (“small Omega”) seem a bit stronger. A function $f(x)$ is said to be $o(g(x))$ if $\alpha g(x)$ grows asymptotically faster than $f(x)$ for any positive α . This implies that $g(x)$ grows much faster than $f(x)$. On the other hand, if $f(x) = \omega(g(x))$, then this implies that $f(x)$ grows much faster than $g(x)$.

From the abstract, [Mal22] seems like it might be a good attempt at providing a through treatment on the concept of big O notation.

The *mean value theorem* says that, for a differentiable function $f(x)$ on the interval $[a, b]$, there exists a point c such that $f'(c)$ is equal to the average rate of change of $f(x)$ on $[a, b]$. Intuitively, if a car takes a trip from Knoxville to Nashville with an average speed of 60 miles per hour, this theorem says that there must be some moment during that trip when the instantaneous speed of the car was 60 miles per hour. This may be represented linearly for a function with potentially multiple variables as

$$\nabla f(c)^\top = \frac{f(x) - f(x_0)}{x - x_0}, \quad c \in [x_0, x].$$

The *chain rule* provides us with a way to think about the derivative of a composite function $h(x) = g(f(x))$ in terms of the derivatives of $f(x)$ and $g(x)$.

In probability, the law of large numbers comes in a weak form and a strong form. In the weak form, it says that the average result of a large number of repetitions of the same experiment (i.e., the *sample mean*) will become arbitrarily close to the expectation or expected value as the number of repetitions approaches infinity. In the strong form, it says that the sample mean will actually converge to the expectation.

Also from probability, *Markov's inequality* places an upper bound on the probability that a random variable X takes on a value of at least a , for some value of a . *Chebyshev's inequality* places an upper bound on the likelihood of a random variable X taking on a value that is at least some distance t away from the expected value for X .

LECTURE 2 OPTIMALITY

We are given an optimization problem and are tasked with finding a *best* decision, where *best* for us typically means global minimum (or perhaps also maximum). So how do we find our solution? Well, let's suppose our problem is one-dimensional; furthermore, let's say our corresponding objective function is $f(x)$. Then one way that we may search for our solution is to check the value of our objective function at fixed intervals; in other words, we may perform a grid search on f over some interval corresponding to the domain of f —say, the interval $[-100, 100]$. Then we may report the smallest value we see as being our solution (assuming of course that we are after the global minimum rather than the maximum).

This is a straightforward approach in one dimension, but the search space explodes as the number of dimensions grows. In the one-dimensional case, we must perform $\mathcal{O}(1/\epsilon)$ evaluations if ϵ is our increment; in the two-dimensional case, the number of evaluations becomes $\mathcal{O}(1/\epsilon^2)$. It turns out that, if n is our number of dimensions, then we will need to perform $\mathcal{O}(1/\epsilon^n)$ evaluations if we wish to proceed in this way.

Let our optimization problem be

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad f(x).$$

We say that $f(x^*)$ is our *global minimum* if $f(x) \geq f(x^*)$ for all $x \in \mathbb{R}^n$. We call $f(x^*)$ a *strict global minimum* if equality holds only whenever $x = x^*$.

The definition of a *local minimum* is a bit more exciting than that of a global minimum. A value $f(x^*)$ may only be a local minimum whenever we are able to construct a *neighborhood* around x^* such that $f(x) \geq f(x^*)$ for every x in this neighborhood. How do we define the neighborhood? All we have to do is choose some $\epsilon > 0$; then, our neighborhood is made up of all those points that are within a distance of ϵ from x^* .

If our objective function is in one, two, or three dimensions, then we can graph it and eyeball where our local or global minima may be. In the case of $f(x) = \|x\|^3$, the global minimum is quite easy to identify; however, it is less obvious that it is more specifically a *strict* global minimum. A similar thing may be said for $f(x) = -\|x\|^3$. However, whether $f(x) = x^3$ has any local or global minima is not quite as obvious. At $x = 0$ it looks as though one might be able to argue for the existence of a local minimum or maximum; however, there isn't one, the reason being because it is impossible to construct a neighborhood around x such that the conditions for a local minimum are met.

There are a couple of conditions that hold any time a point x^* is locally optimal. The first is that the gradient at that point is zero ($\nabla f(x^*) = 0$). The second is that the *Hessian* at that point is *positive semi-definite*: $\nabla^2 f(x^*) \succeq 0$. These conditions hold any time a point is locally optimal, but they do not imply

that a point is a local optimum: they just tell us that it *may* be. The two conditions that, when satisfied, tell us with certainty that a point is locally optimal are as follows: $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succ 0$. It turns out that all we needed was for the Hessian to be *positive definite* rather than positive semi-definite. And how do we check whether a matrix is positive definite, or positive semi-definite for that matter? We check its eigenvalues: a positive definite matrix has all positive eigenvalues; the eigenvalues of a positive semi-definite matrix are all nonnegative.

There are a few reasons why these “optimality conditions” are worth caring about. The first and most obvious of these is that they allow us to check whether a point is or is not locally optimal. If we are interested in determining whether a point $f(x)$ is locally optimal, but we find that $\nabla f(x) \neq 0$, then we know that it can’t be. On the other hand, if we show, for some $f(x)$, that $\nabla f(x) = 0$ and $\nabla^2 f(x) \succ 0$, then we know that we have just found a local minimum. These conditions may also be exploited by algorithms which can use them to determine when they may throw out a point—which was determined not to be locally optimal—or when to return a point—which was found to be a local optimum.

Another sufficient condition for the existence of an optimal solution is given to us by the *Weierstrass theorem*, which tells us that every continuous function has a local optimum whenever its domain is *compact*.

It turns out that similar optimality conditions hold in the quadratic case, where our objective function looks something akin to

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad \frac{1}{2} x^\top Q x + b^\top x + c, \quad Q = Q^\top.$$

Whenever the gradient of this function is zero and the Hessian is positive definite, the function has a unique global minimum, which is given to us by $x^* = -Q^{-1}b$.

[VB04] is a set of lecture notes that has been cited close to 90,000 times! Also, [Bha] is a textbook on positive definite matrices.

Bibliography

- [Bea73] Raymond Beauregard. *A First Course in Linear Algebra*. Houghton Mifflin, 1973.
- [Bha] Rajendra Bhatia. *Positive Definite Matrices*. Princeton University Press.
- [Kun86] Kenneth Kunen. *Set Theory: An Introduction to Independence Proofs*. North Holland, 1986.
- [Mal22] Rouf Ali Firdous Ahmad Mala. The big-o of mathematics and computer science. *Journal of Applied Mathematics and Computation*, 2022.
- [VB04] Lieven Vandenberghe and Stephen Boyd. *Convex optimization*, volume 1. Cambridge university press Cambridge, 2004.